# Exponential Family Distributions in Machine Learning: Complete Reference with Derivations

CS229 Machine Learning Course

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## 1 Introduction

The exponential family is a broad class of probability distributions that forms the foundation of Generalized Linear Models (GLMs) in machine learning. Understanding these distributions and their properties is crucial for building and analyzing many statistical models used in practice.

A distribution belongs to the exponential family if it can be written in the canonical form:

$$p(y;\eta) = b(y)\exp(\eta^T T(y) - a(\eta))$$

where:

- $\eta$ is the **natural parameter** (also called **canonical parameter**)
- $T(y)$ is the **sufficient statistic**
- $a(\eta)$ is the **log partition function**
- $b(y)$ is the **base measure**

## 2 Fundamental Properties

### 2.1 Mean and Variance Relationships

For any exponential family distribution, there exist fundamental relationships between the log partition function and the moments:

**Mean Property:**

$$\mathbb{E}[T(Y);\eta] = \frac{\partial}{\partial\eta}a(\eta)$$

**Variance Property:**

$$\mathrm{Var}(T(Y);\eta) = \frac{\partial^2}{\partial\eta^2}a(\eta)$$

### 2.2 Derivation of Mean Property

Starting with the normalization constraint:

$$\int p(y;\eta)\,dy = 1$$

Taking the derivative with respect to $\eta$:

$$\frac{\partial}{\partial \eta} \int p(y;\eta)\, dy = \frac{\partial}{\partial \eta} 1 = 0$$

Interchanging differentiation and integration:

$$\int \frac{\partial}{\partial \eta} p(y;\eta)\, dy = 0$$

Computing the derivative:

$$\frac{\partial}{\partial \eta} p(y;\eta) = b(y) \exp(\eta^T T(y) - a(\eta)) \left( T(y) - \frac{\partial a(\eta)}{\partial \eta} \right)$$

Substituting back:

$$\int T(y) p(y;\eta)\, dy - \frac{\partial a(\eta)}{\partial \eta} \int p(y;\eta)\, dy = 0$$

Since $\int p(y;\eta)\, dy = 1$:

$$\mathbb{E}[T(Y);\eta] = \frac{\partial a(\eta)}{\partial \eta}$$

# 3 Common Exponential Family Distributions

## 3.1 Bernoulli Distribution

**Standard Form:**

$$p(y;\phi) = \phi^y (1-\phi)^{1-y}, \quad y \in \{0,1\}$$

**Derivation to Exponential Family Form:**

$$\begin{aligned} p(y;\phi) &= \phi^y (1-\phi)^{1-y} && (1)\\ &= \exp(y \log \phi + (1-y)\log(1-\phi)) && (2)\\ &= \exp\left( y \log\left( \frac{\phi}{1-\phi} \right) + \log(1-\phi) \right) && (3) \end{aligned}$$

**Exponential Family Parameters:**

$$\begin{aligned} \eta &= \log\left( \frac{\phi}{1-\phi} \right) \quad \text{(logit function)} && (4)\\ T(y) &= y && (5)\\ a(\eta) &= \log(1+e^\eta) && (6)\\ b(y) &= 1 && (7) \end{aligned}$$

**Mean and Inverse Relationships:**

$$\begin{aligned} \phi &= \frac{1}{1+e^{-\eta}} = \sigma(\eta) \quad \text{(sigmoid function)} && (8)\\ \mathbb{E}[Y] = \phi &= \frac{\partial a(\eta)}{\partial \eta} = \frac{e^\eta}{1+e^\eta} = \sigma(\eta) && (9)\\ \mathrm{Var}(Y) = \phi(1-\phi) &= \frac{\partial^2 a(\eta)}{\partial \eta^2} = \sigma(\eta)(1-\sigma(\eta)) && (10) \end{aligned}$$

**Canonical Response Function:**

$$g(\eta) = \mathbb{E}[T(Y); \eta] = \sigma(\eta) = \frac{1}{1 + e^{-\eta}}$$

**Canonical Link Function:**

$$g^{-1}(\mu) = \log\left(\frac{\mu}{1-\mu}\right) \quad \text{(logit link)}$$

### 3.2 Gaussian Distribution (Known Variance)

**Standard Form:**

$$p(y; \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right)$$

For simplicity, set $\sigma^2 = 1$:

**Derivation to Exponential Family Form:**

$$p(y; \mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(y-\mu)^2}{2}\right) \tag{11}$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2}\right) \exp\left(\mu y - \frac{\mu^2}{2}\right) \tag{12}$$

**Exponential Family Parameters:**

$$\eta = \mu \tag{13}$$

$$T(y) = y \tag{14}$$

$$a(\eta) = \frac{\eta^2}{2} = \frac{\mu^2}{2} \tag{15}$$

$$b(y) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2}\right) \tag{16}$$

**Mean and Variance:**

$$\mathbb{E}[Y] = \mu = \frac{\partial a(\eta)}{\partial \eta} = \eta \tag{17}$$

$$\text{Var}(Y) = \sigma^2 = 1 = \frac{\partial^2 a(\eta)}{\partial \eta^2} \tag{18}$$

**Canonical Response Function:**

$$g(\eta) = \mathbb{E}[T(Y); \eta] = \eta \quad \text{(identity function)}$$

**Canonical Link Function:**

$$g^{-1}(\mu) = \mu \quad \text{(identity link)}$$

### 3.3 Poisson Distribution

**Standard Form:**

$$p(y;\lambda) = \frac{e^{-\lambda}\lambda^y}{y!}, \quad y \in \{0, 1, 2, \ldots\}$$

**Derivation to Exponential Family Form:**

$$p(y;\lambda) = \frac{e^{-\lambda}\lambda^y}{y!} \tag{19}$$

$$= \frac{1}{y!}e^{-\lambda}e^{y\log\lambda} \tag{20}$$

$$= \frac{1}{y!}\exp(y\log\lambda - \lambda) \tag{21}$$

**Exponential Family Parameters:**

$$\eta = \log\lambda \tag{22}$$

$$T(y) = y \tag{23}$$

$$a(\eta) = \lambda = e^{\eta} \tag{24}$$

$$b(y) = \frac{1}{y!} \tag{25}$$

**Mean and Variance:**

$$\lambda = e^{\eta} \tag{26}$$

$$\mathbb{E}[Y] = \lambda = \frac{\partial a(\eta)}{\partial \eta} = e^{\eta} \tag{27}$$

$$\mathrm{Var}(Y) = \lambda = \frac{\partial^2 a(\eta)}{\partial \eta^2} = e^{\eta} \tag{28}$$

**Canonical Response Function:**

$$g(\eta) = \mathbb{E}[T(Y);\eta] = e^{\eta} \quad \text{(exponential function)}$$

**Canonical Link Function:**

$$g^{-1}(\mu) = \log\mu \quad \text{(log link)}$$

### 3.4 Exponential Distribution

**Standard Form:**

$$p(y;\lambda) = \lambda e^{-\lambda y}, \quad y \geq 0$$

**Derivation to Exponential Family Form:**

$$p(y;\lambda) = \lambda e^{-\lambda y} \tag{29}$$

$$= \exp(\log\lambda - \lambda y) \tag{30}$$

$$= \exp((-\lambda)y + \log\lambda) \tag{31}$$

**Exponential Family Parameters:**

$$\eta = -\lambda \tag{32}$$
$$T(y) = y \tag{33}$$
$$a(\eta) = -\log(-\eta) = \log\left(\frac{1}{-\eta}\right) \tag{34}$$
$$b(y) = 1 \tag{35}$$

**Mean and Variance:**

$$\lambda = -\eta \tag{36}$$
$$\mathbb{E}[Y] = \frac{1}{\lambda} = \frac{\partial a(\eta)}{\partial \eta} = \frac{1}{-\eta} \tag{37}$$
$$\mathrm{Var}(Y) = \frac{1}{\lambda^2} = \frac{\partial^2 a(\eta)}{\partial \eta^2} = \frac{1}{\eta^2} \tag{38}$$

**Canonical Response Function:**

$$g(\eta) = \mathbb{E}[T(Y); \eta] = \frac{1}{-\eta}$$

**Canonical Link Function:**

$$g^{-1}(\mu) = -\frac{1}{\mu} \quad \text{(negative inverse link)}$$

### 3.5 Gamma Distribution

**Standard Form (shape $\alpha$, rate $\beta$):**

$$p(y; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y}, \quad y > 0$$

For fixed shape parameter $\alpha$, varying only $\beta$:
**Derivation to Exponential Family Form:**

$$p(y; \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y} \tag{39}$$
$$= \frac{y^{\alpha-1}}{\Gamma(\alpha)} \exp(\alpha \log \beta - \beta y) \tag{40}$$
$$= \frac{y^{\alpha-1}}{\Gamma(\alpha)} \exp((-\beta) y + \alpha \log \beta) \tag{41}$$

**Exponential Family Parameters:**

$$\eta = -\beta \tag{42}$$
$$T(y) = y \tag{43}$$
$$a(\eta) = -\alpha \log(-\eta) = \alpha \log\left(\frac{1}{-\eta}\right) \tag{44}$$
$$b(y) = \frac{y^{\alpha-1}}{\Gamma(\alpha)} \tag{45}$$

**Mean and Variance:**

$$\mathbb{E}[Y] = \frac{\alpha}{\beta} = \frac{\partial a(\eta)}{\partial \eta} = \frac{\alpha}{-\eta} \tag{46}$$

$$\mathrm{Var}(Y) = \frac{\alpha}{\beta^2} = \frac{\partial^2 a(\eta)}{\partial \eta^2} = \frac{\alpha}{\eta^2} \tag{47}$$

**Canonical Response Function:**

$$g(\eta) = \mathbb{E}[T(Y); \eta] = \frac{\alpha}{-\eta}$$

**Canonical Link Function:**

$$g^{-1}(\mu) = -\frac{\alpha}{\mu} \quad \text{(negative inverse link)}$$

### 3.6 Multinomial Distribution (Categorical)

**Standard Form:**

$$p(y; \phi_1, \ldots, \phi_k) = \prod_{j=1}^{k} \phi_j^{y_j}$$

where $y = (y_1, \ldots, y_k)$ with $y_j \in \{0, 1\}$, $\sum_{j=1}^{k} y_j = 1$, and $\sum_{j=1}^{k} \phi_j = 1$.
For $k$ classes, we use $k - 1$ parameters (the last probability is determined by the constraint).
**Derivation to Exponential Family Form:**

$$p(y; \phi) = \prod_{j=1}^{k} \phi_j^{y_j} \tag{48}$$

$$= \prod_{j=1}^{k-1} \phi_j^{y_j} \cdot \phi_k^{y_k} \tag{49}$$

$$= \prod_{j=1}^{k-1} \phi_j^{y_j} \cdot \phi_k^{1-\sum_{j=1}^{k-1} y_j} \tag{50}$$

$$= \phi_k \prod_{j=1}^{k-1} \left(\frac{\phi_j}{\phi_k}\right)^{y_j} \tag{51}$$

$$= \exp\left(\sum_{j=1}^{k-1} y_j \log\left(\frac{\phi_j}{\phi_k}\right) + \log \phi_k\right) \tag{52}$$

**Exponential Family Parameters:**

$$\eta_j = \log\left(\frac{\phi_j}{\phi_k}\right) \quad j = 1, \ldots, k-1 \tag{53}$$

$$T(y) = (y_1, \ldots, y_{k-1})^T \tag{54}$$

$$a(\eta) = \log\left(1 + \sum_{j=1}^{k-1} e^{\eta_j}\right) \tag{55}$$

$$b(y) = 1 \tag{56}$$

**Inverse Relationships:**

$$\phi_j = \frac{e^{\eta_j}}{1 + \sum_{\ell=1}^{k-1} e^{\eta_\ell}} \quad j = 1, \ldots, k-1 \tag{57}$$

$$\phi_k = \frac{1}{1 + \sum_{\ell=1}^{k-1} e^{\eta_\ell}} \tag{58}$$

**Canonical Response Function:**

$$g_j(\eta) = \mathbb{E}[y_j] = \phi_j = \frac{e^{\eta_j}}{1 + \sum_{\ell=1}^{k-1} e^{\eta_\ell}} \quad \text{(softmax function)}$$

**Canonical Link Function:**

$$g_j^{-1}(\mu_j) = \log\left(\frac{\mu_j}{\mu_k}\right) \quad \text{(multinomial logit link)}$$

# 4 Summary Table of Exponential Family Distributions

| **Distribution** | **Natural Parameter** $\eta$ | **Sufficient Statistic** $T(y)$ | **Log Partition** $a(\eta)$ | **Mean** $\mathbb{E}[T(Y)]$ |
|---|---|---|---|---|
| Bernoulli | $\log\left(\frac{\phi}{1-\phi}\right)$ | $y$ | $\log(1+e^{\eta})$ | $\frac{1}{1+e^{-\eta}}$ |
| Gaussian | $\mu$ | $y$ | $\frac{\eta^2}{2}$ | $\eta$ |
| Poisson | $\log\lambda$ | $y$ | $e^{\eta}$ | $e^{\eta}$ |
| Exponential | $-\lambda$ | $y$ | $-\log(-\eta)$ | $\frac{1}{-\eta}$ |
| Gamma | $-\beta$ | $y$ | $-\alpha\log(-\eta)$ | $\frac{\alpha}{-\eta}$ |

# 5 Response Functions and Link Functions

| **Distribution** | **Canonical Response Function** $g(\eta)$ | **Canonical Link Function** $g^{-1}(\mu)$ |
|---|---|---|
| Bernoulli | $\frac{1}{1+e^{-\eta}}$ (Sigmoid) | $\log\left(\frac{\mu}{1-\mu}\right)$ (Logit) |
| Gaussian | $\eta$ (Identity) | $\mu$ (Identity) |
| Poisson | $e^{\eta}$ (Exponential) | $\log\mu$ (Log) |
| Exponential | $\frac{1}{-\eta}$ (Negative Inverse) | $-\frac{1}{\mu}$ (Negative Inverse) |
| Gamma | $\frac{\alpha}{-\eta}$ (Scaled Negative Inverse) | $-\frac{\alpha}{\mu}$ (Scaled Negative Inverse) |
| Multinomial | $\frac{e^{\eta_j}}{1+\sum_\ell e^{\eta_\ell}}$ (Softmax) | $\log\left(\frac{\mu_j}{\mu_k}\right)$ (Multinomial Logit) |

# 6 Generalized Linear Models (GLMs)

Exponential family distributions form the foundation of Generalized Linear Models. A GLM makes three key assumptions:

1. $y|x;\theta \sim \text{ExponentialFamily}(\eta)$
2. The goal is to predict $\mathbb{E}[T(y)|x;\theta]$
3. The natural parameter is linearly related to input features: $\eta = \theta^T x$

### 6.1 Linear Regression as GLM

For Gaussian distribution with identity link:

$$h_\theta(x) = \mathbb{E}[y|x;\theta] = \mu = \eta = \theta^T x \tag{59}$$

### 6.2 Logistic Regression as GLM

For Bernoulli distribution with logit link:

$$h_\theta(x) = \mathbb{E}[y|x;\theta] = \phi = \frac{1}{1+e^{-\eta}} = \frac{1}{1+e^{-\theta^T x}} \tag{60}$$

### 6.3 Poisson Regression as GLM

For Poisson distribution with log link:

$$h_\theta(x) = \mathbb{E}[y|x;\theta] = \lambda = e^\eta = e^{\theta^T x} \tag{61}$$

## 7 Maximum Likelihood Estimation

For a dataset $\{(x^{(i)}, y^{(i)})\}_{i=1}^m$, the log-likelihood for an exponential family GLM is:

$$\ell(\theta) = \sum_{i=1}^m \log p(y^{(i)}|x^{(i)};\theta) = \sum_{i=1}^m \left[\eta^{(i)} T(y^{(i)}) - a(\eta^{(i)}) + \log b(y^{(i)})\right]$$

where $\eta^{(i)} = \theta^T x^{(i)}$.

### 7.1 Gradient of Log-Likelihood

The gradient with respect to $\theta$ is:

$$\frac{\partial \ell(\theta)}{\partial \theta} = \sum_{i=1}^m \left(T(y^{(i)}) - \frac{\partial a(\eta^{(i)})}{\partial \eta^{(i)}}\right) x^{(i)} = \sum_{i=1}^m (T(y^{(i)}) - \mathbb{E}[T(Y^{(i)})|x^{(i)};\theta]) x^{(i)}$$

This leads to the general update rule:

$$\theta := \theta + \alpha \sum_{i=1}^m (T(y^{(i)}) - h_\theta(x^{(i)})) x^{(i)}$$

## 8 Convexity Properties

The negative log-likelihood of exponential family GLMs is convex. This follows from the fact that:

$$\frac{\partial^2 a(\eta)}{\partial \eta^2} = \mathrm{Var}(T(Y);\eta) \geq 0$$

The Hessian of the negative log-likelihood is:

$$H = \sum_{i=1}^m \frac{\partial^2 a(\eta^{(i)})}{\partial \eta^{(i)} \partial \eta^{(i)}} x^{(i)} (x^{(i)})^T$$

Since $\frac{\partial^2 a(\eta)}{\partial \eta^2} \geq 0$ (being a variance), the Hessian is positive semi-definite, ensuring convexity.

# 9 Conclusion

Exponential family distributions provide a unified framework for understanding many common probability distributions used in machine learning. Their elegant mathematical structure, with natural parameters, sufficient statistics, and log partition functions, enables:

- Unified treatment of regression and classification problems
- Guaranteed convex optimization for maximum likelihood estimation
- Natural interpretation of link functions and response functions
- Principled approach to model selection and comparison

Understanding these distributions and their properties is fundamental to mastering modern statistical machine learning methods, from simple linear and logistic regression to more complex generalized linear models and beyond.